# Unlearn in Context: Contrastive Alignment via Diversified Supervision for Diffusion Models

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## Abstract

*Concept unlearning has emerged as a promising direction for reducing the risks of harmful content generation in text-to-image diffusion models by selectively erasing undesirable concepts from a model's parameters. Existing approaches typically rely on keywords to identify the target concept to be unlearned. However, we show that this keyword-based formulation is inherently limited: a visual concept is multi-dimensional, can be expressed in diverse textual forms, and often overlaps with related concepts in the latent space. We therefore adopt a contextual view: a concept should be unlearned through the company it keeps. Under this view, keyword-based unlearning approaches, which imprecisely indicate the target concept via a fixed representation, are brittle against prompt-rephrasing jailbreaks and highly prone to collateral over-forgetting. To address these limitations, we propose* ***Diversified Unlearning****, a framework that exposes the erasure objective to natural context-dependent embedding variations. Rather than relying on simple prompt augmentation, we construct paired contextual formulations that introduce* ***implicit contrastive supervision*** *directly into the forgetting process, encouraging the model to isolate the target concept while keeping surrounding contexts invariant. Through extensive experiments across multiple benchmarks and state-of-the-art baselines, we demonstrate that integrating our framework as a lightweight add-on into existing unlearning pipelines yields notable improvements in both erasure effectiveness and concept retention, while significantly hardening models against adversarial recovery attacks. All experimental results and detailed implementations can be found at* https://vnpt-ai-official.github.io/WACV_diversified_unlearning/.

## 1. Introduction

Recent text-to-image generative models, such as Stable Diffusion [31] or Dall-E [29], inadvertently learn **undesirable concepts** including *toxicity, social biases, and copyrighted material* from web-scale data [33]. Consequently, these models can be exploited to generate harmful or infringing content [6, 37], posing severe societal risks. While data filtering before training (pre-processing) and output censorship (post-processing) mitigate these risks, both remain limited by high computational costs and inefficiency under continuous large-scale queries.

A more promising direction is ***concept unlearning***, which directly removes undesired concepts from a model by fine-tuning its parameters [4, 5, 10, 11, 22, 40]. Existing approaches generally fall into two categories: *Output-based methods* [5, 10] that force the noise prediction of a target concept $c_e$ to match a neutral anchor $c_a$, and *Attention-based methods* [11, 22] that modify cross-attention weights to weaken text-to-visual alignments.

Despite their intuitive appeal, both families are typically formulated in a ***keyword-based*** manner, representing a visual concept by isolated textual tokens (e.g., "Barack Obama" or "a photo of Barack Obama"). However, visual concepts are not tied to a single lexical form: the same target can appear through paraphrases, attributes, actions, and surrounding scenes. This limitation echoes a classic distributional principle in linguistics: "You shall know a word by the company it keeps" [9]. We adapt this principle to concept unlearning: *one shall unlearn a word by the company it keeps*. In text-to-image models, this "company" consists of the natural contexts in which a concept appears, including its attributes, actions, co-occurring objects, backgrounds, and paraphrased textual descriptions. Reliable unlearning should therefore identify the target across these contextual variations while preserving the surrounding context itself. Through systematic analysis, we identify two failure modes showing why keyword-based unlearning falls short of this contextual principle:

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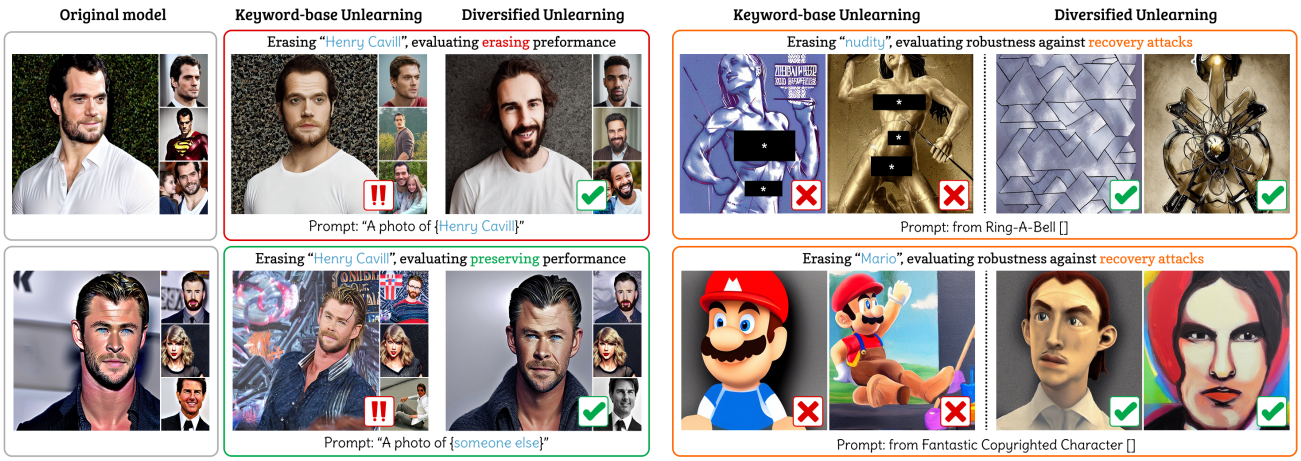



Figure 1. Diversified Unlearning enhances the representation of target concepts on top of state-of-the-art methods, significantly improves both erasing performance on target concepts and preserving abilities on unrelated ones. Moreover, our method also demonstrated effectiveness in mitigating recovery attacks against sexual concepts and copyrighted characters. (✓ good performance; ✗ bad performance; !! moderate performance). (*) Censoring for publication.

- **Observation 1: Imprecise Target Representation.** During pre-training, concepts are entangled with diverse linguistic contexts, forming context-dependent representation regions rather than isolated points. Our perturbation analysis (Table 1) shows that, even after adding noise to the text embedding, the underlying diffusion model can still generate the target concept $c_e$ with high probability. This indicates that the concept remains recoverable from a neighborhood of semantically related embeddings. Keyword-based unlearning, which uses a single fixed representation to identify the target, may therefore suppress the canonical keyword while leaving nearby contextual variants untouched, making the model susceptible to **prompt rephrasing jailbreaks** and noise-based recovery attacks (Table 5).
- **Observation 2: Ambiguous Erasure Signals in Output Space.** In output-based unlearning, a prompt containing an isolated keyword $c_e$ is not decoded in isolation: the stochastic reverse process often also produces co-occurring objects, attributes, and background contexts in intermediate latents. Forcing the full prediction trajectory to match a generic anchor $c_a$ provides an ambiguous training signal because the loss does not specify which visual factors correspond to the target and which should be preserved. As a result, benign contextual features can be suppressed together with the target. This issue is further exacerbated by traditional data augmentation or adversarial unlearning strategies [1, 17, 47]; when diversified or adversarially perturbed inputs are mapped to an unconstrained anchor, they can introduce an **overly aggressive erasure signal** that compromises preservation (Table 2).

Motivated by this contextual view, we propose **Diversified Unlearning**, a framework that unlearns a concept through the company it keeps: diverse contextual variations whose non-target content is structurally controlled. Unlike the diversification strategies in Observation 2, where the added context is erased together with the target, the contexts we use to diversify the erasure input are themselves protected from erasure. This design introduces **Implicit Contrastive Supervision**: the model is encouraged to change the target concept while preserving the context shared by the target and anchor pair.

For output-based methods, we introduce **Diversified Prompting**. Instead of mapping an isolated keyword $c_e$ to a generic anchor $c_a$, we construct paired formulations: $\{c + c_e\} \rightarrow \{c + c_a\}$, where $c \sim \mathcal{C}$ denotes clean, natural contexts. Because the context $c$ is shared across both sides of the transformation, the erasure direction is concentrated on the difference between $c_e$ and $c_a$. This context-invariant formulation helps the model distinguish *what to erase* from *what to preserve*, reducing collateral damage to adjacent concepts and background contexts.

For attention-based methods, direct context manipulation can cause severe over-forgetting due to text encoder semantic bias. Because the encoder aggregates semantics from both the target and the context, suppressing the full prompt embedding indiscriminately erases *shared contextual features*, harming general generative ability. We thus propose **Diversified Embedding Mixup**, a ***token-level interpolation*** strategy that blends target tokens with contextualized embeddings. Crucially, these blended contexts are explicitly incorporated into the preservation objective, encouraging the model to isolate the target identity while safeguarding surrounding contextual features against representation collapse.

Our modules realize the same principle in the space where each baseline operates: the context that diversifies the erasure side is also what preservation keeps intact, by

construction in the output space and through an explicit preservation term in the attention space. As lightweight add-ons, they enhance existing unlearning methods by providing richer textual coverage while mitigating the instability and over-forgetting issues. Evaluation across five diverse settings–celebrities, physical objects, copyrighted characters, nudity, and artistic styles (Table 1, Supplementary Material)–demonstrates substantial improvements in both erasure and preservation. For instance, in celebrity erasure (Table 3), our framework achieves a $\sim$**50% reduction** in GPT-Score [25]. More importantly, our approach exhibits strong resilience against advanced adversarial recovery attacks, including Ring-A-Bell [39], Fantastic Copyrighted Concepts [14], and Noise-based Attacks [21]. This structural **robustness**, alongside improved concept **erasure** and **preservation**, manifests as the **triple benefits** highlighted in Figure 1, where upgrading baselines with our method suppresses the regeneration of sensitive targets like "nudity" and "Mario".

In summary, our main contributions are:

① We identify two key limitations of keyword-based concept unlearning: isolated keywords provide imprecise target representations, and output-space objectives can produce ambiguous erasure signals that compromise preservation. We propose **Diversified Unlearning**, a context-structured framework inspired by the principle that a concept should be unlearned through the company it keeps, introducing *implicit contrastive supervision* through paired contextual prompts and token-level embedding mixup.

② Through extensive experiments across diverse evaluation settings, we demonstrate notable improvements in **both erasure effectiveness and concept preservation**. Under adversarial recovery attacks, Diversified Unlearning also exhibits **significantly stronger robustness**, suggesting a valuable perspective on how visual concepts should be represented within the unlearning pipeline.

## 2. Preliminaries and Related Work

**Latent Diffusion Models (LDMs).** Text-conditioned LDMs [16, 29–31] generate images in a latent space compressed by a pretrained encoder $z_0 = \mathcal{E}(x)$. Given a prompt $c$ encoded as $\tau(c)$ by a text encoder such as CLIP [28], the denoising network $\epsilon_\theta$ is trained to predict noise $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ at timestep $t$:

$$\min_{\theta} \mathbb{E}_{z_0,t,\epsilon} \left\| \epsilon - \epsilon_\theta(z_t, t, \tau(c)) \right\|_2^2 . \tag{1}$$

**Concept Unlearning.** Concept erasing aims to update model weights $\theta$ to a sanitized state $\theta'$ that suppresses undesirable concepts $c_e \in \mathbf{E}$ while preserving benign concepts $c_p \in \mathbf{P}$. Existing safety interventions can be grouped into inference-time and finetuning-based methods. Inference-time approaches do not permanently modify model weights: they include post-hoc filtering with detectors such as Nudenet [27], prompt rewriting or negative prompting [14, 35], and latent/attention interventions such as SLD [33], TRCE [32], SAFREE [44]. These methods are flexible, but remain vulnerable to adversarial prompting [3, 43], can be bypassed in open deployments [36], and must be applied at inference time.

Finetuning-based unlearning instead modifies the model parameters and redistributes a sanitized checkpoint. Dataset-level filtering before pre-training [34, 35, 38] is expensive and can leave residual unsafe concepts [10]; parameter-level finetuning is therefore the dominant practical alternative. Current methods broadly follow two paradigms [5]:

*1) Output-based Unlearning:* Methods such as ESD [10], SalUn [8], AP [4], AGE [5], and ACE [40] steer the noise prediction of an erased concept $c_e$ toward a neutral anchor $c_a$, while using a preservation loss to retain benign concepts $c_p$:

$$\begin{aligned} \min_{\theta'} \mathbb{E}_{c_e,c_p} \Big[ & \left\| \epsilon_{\theta'}(z_t, t, \tau(c_e)) - \epsilon_\theta(z_t, t, \tau(c_a)) \right\|_2^2 + \\ & \left\| \epsilon_{\theta'}(z_t, t, \tau(c_p)) - \epsilon_\theta(z_t, t, \tau(c_p)) \right\|_2^2 \Big]. \end{aligned} \tag{2}$$

*2) Attention-based Unlearning:* To avoid sampling noisy latent trajectories, methods such as UCE [11] and MACE [22] directly edit cross-attention projection weights $W$ (e.g., $W^K$ or $W^V$), weakening the alignment between target text features and visual tokens [11, 19, 24, 45]:

$$\min_{W'} \sum_{c_e \in \mathbf{E}} \left\| W'\tau(c_e) - W\tau(c_a) \right\|_2^2 + \sum_{c_p \in \mathbf{P}} \left\| W'\tau(c_p) - W\tau(c_p) \right\|_2^2 . \tag{3}$$

where $W$ and $W'$ denote the original and updated cross-attention weights. This formulation can admit closed-form updates in specific settings [24] and operates purely in embedding space.

**Robust and Diversified Unlearning.** Recent works augment keyword-based erasure with adversarial training, distribution alignment, or data diversification [12, 17, 18, 41, 47, 49], partly motivated by black-box [39, 42] and white-box recovery attacks [7, 48]. However, unconstrained diversification can produce overly aggressive erasure signals that harm preservation; our framework instead uses structured context pairs to separate target removal from context preservation without architectural overhead. Due to space constraints, we provide an extended discussion of recent advancements in the Supplementary Material A.

## 3. An Empirical Analysis of Keyword-based Unlearning

Keyword-based unlearning operationalizes concept erasure by mapping isolated tokens (e.g., $c_e =$ *"Barack Obama"*) to a neutral anchor $c_a$. However, this fixed-indicator formulation only partially reflects the contextual company through

Table 1. Embedding perturbation analysis under varying Gaussian noise ratios $\alpha$ added to the prompt *"a photo of Cassette Player/Church"*. DS-1 and DS-5 denote the top-1 and top-5 detection score of a pre-trained ResNet50 [13].

| SD | *"**Cassette Player**" concept* | | | | | | *"**Church**" concept* | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $\alpha = 0$ | 0.1 | 0.3 | 0.5 | 0.8 | 1.0 | $\alpha = 0$ | 0.1 | 0.3 | 0.5 | 0.8 | 1.0 |
| DS-1 | 5 | 6 | 3 | 3 | 1 | 0 | 86 | 80 | 75 | 38 | 1 | 1 |
| DS-5 | 97 | 100 | 74 | 37 | 8 | 0 | 100 | 100 | 91 | 54 | 4 | 2 |

which concepts are represented in text-to-image (T2I) models, where a target may appear across paraphrases, attributes, actions, and surrounding scenes. In this section, we expand upon the two core observations from Section 1 with empirical evidence suggesting why such erasure can lead to fragile forgetting and suboptimal preservation.

***Observation 1: Imprecise Target Representation.*** During T2I pre-training, visual concepts are learned within diverse linguistic contexts. Consequently, within the text encoder's output space, the embedding of a target concept $c_e$ can vary with surrounding tokens and prompt structure. Keyword-based unlearning, however, optimizes parameters against an isolated keyword, using a single fixed text embedding as the target indicator. This creates a potential mismatch with the context-rich nature of pre-training data.

To probe whether concept information is confined to the canonical text embedding, we inject Gaussian noise (ratio $\alpha$) into the text embedding of *"a photo of Cassette Player/Church"*. As shown in Table 1, even under moderate noise ($\alpha = 0.3$), the model still generates the target concepts at non-trivial rates, with top-5 detection scores (DS-5) of $74\%$ for *"Cassette Player"* and $91\%$ for *"Church"*. This suggests that concept-bearing signals persist in neighborhoods around the canonical embedding. Accordingly, a keyword-only update may suppress the exact training indicator while leaving nearby contextual or perturbed variants insufficiently constrained. This interpretation is consistent with the robustness results in Table 5: standard AP [4] remains more vulnerable than its diversified counterpart under Ring-A-Bell [39] and Noise ATK [21], including a high residual GPT-Score of 39.63 under Noise ATK compared with 14.94 after diversification. These results support the need to cover contextual variants rather than treating the keyword as a single point estimate.

***Observation 2: Ambiguous Erasure Signals in Output Space***. In output-based unlearning, a prompt containing an isolated keyword $c_e$ is not decoded in isolation. During the reverse process, the generated trajectory can also include backgrounds, attributes, or objects associated with the target. For example, when erasing *"Barack Obama"*, intermediate outputs may include co-occurring elements such as a business suit or the American flag. Forcing these complete, multi-object trajectories to match a generic anchor $c_a$ can make the training signal ambiguous: the objective does not explicitly specify which visual factors correspond to the target identity and which should remain preserved. As a result, the erasure signal may also affect surrounding attributes. This interpretation is consistent with the preservation degradation observed in Table 3; for instance, ESD [10] and AP [4] retain relatively low context-preservation GPT-Scores [25] of 31.68 and 14.97, respectively.

Table 2. Object erasure performance across prompt complexities (Supp. Material B.1.2). Erasure Success Rate (ESR) and Preservation Success Rate (PSR) are computed at Top-1 (-1) and Top-5 (-5) (higher is better). Aug-ESD/AP and Adv denote ESD [10]/AP [4] with prompt augmentation and AdvUnlearn [47], respectively.

| Object | *"**Garbage Truck**" erasure* | | | | *"**Five objects**" erasure* | | | |
|---|---|---|---|---|---|---|---|---|
| | ESR-1 ↑ | ESR-5 ↑ | PSR-1 ↑ | PSR-5 ↑ | ESR-1 ↑ | ESR-5 ↑ | PSR-1 ↑ | PSR-5 ↑ |
| ESD | 0.968 | 0.934 | 0.652 | 0.831 | 0.891 | 0.754 | 0.470 | 0.634 |
| Aug-ESD | **1.000** | **1.000** | 0.566 | 0.729 | **0.987** | **0.958** | 0.350 | 0.484 |
| Diversified-ESD | 0.998 | 0.996 | **0.775** | **0.933** | 0.954 | 0.821 | **0.730** | **0.912** |
| AP | **1.000** | 0.998 | 0.675 | 0.864 | 0.922 | 0.829 | 0.606 | 0.780 |
| Aug-AP | **1.000** | **1.000** | 0.613 | 0.799 | **0.982** | **0.957** | 0.454 | 0.636 |
| Diversified-AP | 0.994 | 0.970 | **0.784** | **0.943** | 0.942 | 0.836 | **0.741** | **0.922** |
| Adv | **1.000** | **0.992** | 0.768 | 0.936 | **1.000** | **0.990** | 0.459 | 0.604 |
| Diversified-Adv | 0.996 | 0.976 | **0.807** | **0.952** | 0.988 | 0.974 | **0.732** | **0.916** |

This issue can become more pronounced when prompt augmentation or adversarial unlearning strategies diversify inputs without preserving the associated context structure [1, 17, 47]. Although methods like AP [4] search for preservation concepts, combining them naively with erasure augmentation still degrades retention. Mapping context-enriched targets or adversarial perturbations to isolated anchors forces the loss to suppress co-occurring semantics, while uncoupled preservation sets fail to counteract this over-erasure. Consequently, as shown in Table 2, such one-sided augmentation and adversarial perturbations in Aug-ESD, Aug-AP, and AdvUnlearn [47] causes *sharply dropping multi-object PSR*. This demonstrates that arbitrary data expansion cannot replace structured, contrastive alignment.

## 4. Diversified Unlearning

To overcome the fragility of context-free semantic formulations, we propose **Diversified Unlearning**. Instead of viewing a concept through an isolated keyword, our approach operationalizes that a concept should be unlearned through the company it keeps: erasure is modeled as a structured transformation over diverse contextual variations. By pairing target concepts with natural contextual expressions, we introduce an *implicit contrastive mechanism* that encourages the model to distinguish between the core identity to be erased and the surrounding context to be preserved. Here, *contrastive* refers to paired context-invariant alignment rather than an additional contrastive-loss objective.

***Diversified Prompting.*** To resolve the limitations of keyword-based erasure, we propose **Diversified Prompting**, which replaces the unconstrained keyword objective with a context-invariant formulation. For each target con-

cept $c_e$, we sample natural contexts $c \sim \mathbf{C}$ to form composite pairs (e.g., pairing $c_e$ "Henry Cavill" with $c$ "waving" to yield "Henry Cavill waving"). The output-space erasure loss is optimized over these contextualized formulations:

$$\min_{\theta'} \mathbb{E}_{c_e \in \mathbf{E}, c_p \in \mathbf{P}, c \in \mathbf{C}} \Bigg[ \underbrace{\|\epsilon_{\theta'}(\tau(c + c_e)) - \epsilon_\theta(\tau(c + c_a))\|_2^2}_{L_1} + \delta \cdot \underbrace{\|\epsilon_{\theta'}(\tau(c_p)) - \epsilon_\theta(\tau(c_p))\|_2^2}_{L_2} \Bigg], \tag{4}$$

where $\delta \in \{0, 1\}$ activates preservation loss $L_2$ only if required by the baseline; $z_t$ and $t$ are omitted for simplicity.

This architecture is designed to address the two limitations outlined in our empirical analysis. On the one hand, by keeping the surrounding context $c$ invariant across both sides of $L_1$, the optimization provides a structural alignment that helps *distinguish what to erase from what to preserve*; it encourages target removal while reducing unintended changes to co-occurring background content, mitigating the output-space ambiguity described in **Observation 2**. On the other hand, embedding $c_e$ within natural phrases *aligns with the text encoder's pre-training distribution*, exposing the optimization to context-sensitive variations that can improve robustness to prompt rephrasing jailbreaks (**Observation 1**).

***Diversified Embedding Mixup.*** While Diversified Prompting works well in the **output-based setting**, applying the same strategy directly to **attention-based unlearning** often causes *severe over-forgetting*. This stems from the **semantic bias of the text encoder**: even non-sensible prompts (e.g., an empty string "") are mapped to meaningful embeddings that yield coherent outputs. In output-based unlearning (Equation (2)), the loss is *bounded by a realistic output distribution*, preventing collapse. However, in attention-based unlearning (Equation (3)), *no such regularization exists*, often leading to *non-sensible generations* for erased concepts, a phenomenon also noted by [11].

To address this, we introduce **diversification directly in the embedding space**. Specifically, we define a mixup function $f(\tau(c_e), \mathbf{C})$ that interpolates the token embedding of $c_e$ with *contextualized embeddings* $c$. The diversified attention-based loss is:

$$\min_{W'} \sum_{c_e \in \mathbf{E}} \left\| W' f(\tau(c_e), \mathbf{C}) - W\tau(c_a) \right\|_2^2 + \sum_{c_p \in \mathbf{P} \cup \mathbf{C}} \left\| W' \tau(c_p) - W\tau(c_p) \right\|_2^2, \tag{5}$$

where the **mixup function** is defined *token-wise*. Let $\mathcal{I}(c_e)$ denote the token positions corresponding to the target concept in the tokenized input. In the mixed branch, $\tau(c)^i$ denotes the contextual-concept embedding aligned to the corresponding target-token slot:

$$f(\tau(c_e), \mathbf{C})^i = \begin{cases} \tau(c_e)^i, & \text{if } i \notin \mathcal{I}(c_e) \\ \dfrac{1}{|\mathbf{C}|} \displaystyle\sum_{c \in \mathbf{C}} \Big( \alpha \tau(c_e)^i + (1-\alpha)\tau(c)^i \Big), & \text{if } i \in \mathcal{I}(c_e). \end{cases} \tag{6}$$

For example, if the input prompt after tokenization is represented as "⟨SOS⟩, ⟨a⟩, ⟨photo⟩, ⟨of⟩, ⟨Henry⟩, ⟨Cavill⟩, ⟨PAD⟩, ..." where the tokens ⟨Henry⟩ and ⟨Cavill⟩ correspond to the **concept** $c_e$ and the remaining tokens are treated as *template tokens*, we replace the embeddings of $c_e$ with mixed features using *contextual concepts* $c \in \mathbf{C}$ (e.g., "a man", "an actor"). Unlike the simple *prompt-level mixup* $\alpha\,\tau(c_e) + (1-\alpha)\,\tau(c)$, this **token-level formulation** yields better performance.

This design *preserves the identity of the target concept* while injecting *contextual diversity*. In practice, we set $\alpha$ close to 1, retaining most of the target signal while adding just enough variation to *mitigate over-forgetting*. As a lightweight add-on, Diversified Embedding Mixup applies preservation regularization over $\mathbf{P}$ only when present in the baseline objective. Since contextual concepts $c \in \mathbf{C}$ are mixed with the target concept $c_e$, we also *preserve* $c$ to prevent unintended forgetting, which motivates careful selection of contextual concepts.

**Context Set Construction.** To implement Diversified Unlearning, we construct a context set $\mathbf{C}$ for each target concept $c_e$ using advanced LLMs (e.g., GPT-4o/GPT-5), aiming to approximate how the concept **naturally appears in real-world usage**. We employ a **unified prompt-design rule** and provide a single illustrative example per setting to prompt the LLMs (see Supplementary Material B).

Applied exclusively to training prompts, our filtering pipeline uses a task-aligned recognition model on synthesized images. Our generation rules guarantee high-confidence detections for every candidate, bypassing the need for hyperparameter tuning. For celebrity erasure, we expand simple templates with everyday actions (e.g., "a photo of Henry Cavill smiling"); for artistic styles, we vary the subject while anchoring the style descriptor (e.g., "an art of a fox in the style of Van Gogh"); and for explicit concepts, we blend the keyword with diverse subjects (e.g., "nudity portrait"). This principled construction prevents optimization from overfitting to a fixed keyword token, directly enhancing *robustness and generality* of erasure.

## 5. Experiments

In this section, we present a **comprehensive evaluation** of our method's ability to enhance existing unlearning approaches, effectively *erasing diverse concepts* while *preserving essential knowledge*. We benchmark our Diversified Unlearning against five recent and representative era-

Table 3. Quantitative comparison between keyword baselines and our diversified variants on ten-celebrity erasure.

| Method | Erasure | | Preservation | | | |
|---|---|---|---|---|---|---|
| | CLIP-i↓ | GPT↓ | LPIPS↓ | CLIP-i↑ | CLIP-t↑ | GPT↑ |
| ESD | 65.98 | 11.78 | 0.66 | 62.59 | 24.25 | 31.68 |
| Diversified-ESD | **62.58** | **5.50** | **0.63** | **69.73** | **26.28** | **46.15** |
| UCE | 60.59 | 4.48 | 0.65 | 62.90 | 23.97 | **31.42** |
| Diversified-UCE | **58.55** | **2.83** | **0.64** | **66.92** | **24.57** | 24.40 |
| AP | 61.14 | 13.55 | 0.79 | 49.31 | 21.34 | 14.97 |
| Diversified-AP | **60.78** | **6.10** | **0.68** | **70.50** | **27.09** | **51.15** |
| AGE | 60.68 | 18.84 | 0.83 | 42.31 | 17.96 | 6.98 |
| Diversified-AGE | **58.84** | **12.59** | **0.71** | **64.20** | **26.15** | **12.59** |
| ACE | 74.69 | 29.45 | 0.46 | 84.64 | **29.92** | **76.72** |
| Diversified-ACE | **68.67** | **13.53** | **0.39** | **85.91** | 28.88 | 75.05 |

Table 4. Quantitative comparison between keyword baselines and our diversified variants on "Mario" erasure.

| Method | Erasure | | Preservation | | | |
|---|---|---|---|---|---|---|
| | CLIP-i↓ | GPT↓ | LPIPS↓ | CLIP-i↑ | CLIP-t↑ | GPT↑ |
| ESD | 66.38 | 16.25 | 0.34 | 91.66 | 31.54 | 72.18 |
| Diversified-ESD | **66.29** | **13.50** | **0.32** | **92.38** | **31.61** | **73.65** |
| UCE | 70.23 | **21.78** | 0.43 | 93.92 | **31.95** | 94.78 |
| Diversified-UCE | **71.97** | 26.58 | **0.37** | **94.98** | 31.93 | **94.83** |
| AP | 64.89 | 12.70 | 0.33 | 92.50 | 31.54 | 73.95 |
| Diversified-AP | **63.42** | **8.60** | **0.31** | **92.56** | **31.73** | **74.35** |
| AGE | 62.49 | 11.03 | 0.35 | 90.62 | 31.53 | 70.83 |
| Diversified-AGE | **60.29** | **6.00** | **0.34** | **90.87** | **31.53** | **71.05** |
| ACE | 64.52 | **0.28** | 0.30 | 91.50 | 31.84 | 89.15 |
| Diversified-ACE | **63.55** | 1.05 | **0.27** | **93.12** | **32.02** | **91.23** |

sure techniques: ESD [10], UCE [11], AP [4], AGE [5], and ACE [40] under identical experimental settings. In total, we evaluate **10 unlearning methods** across **five target scenarios**, evaluating *erasure fidelity*, *benign concept preservation*, and *robustness against recovery attacks*. Across 50 erasure/preservation comparisons, diversified variants achieve **37 wins, 6 ties, and 7 losses**; the complete per-setting breakdown and W/T/L assignment rule are provided in Table 1 of the Supplementary Material B.1. The empirical results demonstrate that our framework yields systematic advantages over keyword-based counterparts across these three evaluation aspects. Due to page limits, quantitative analysis for three representative settings—*celebrity*, *copyrighted*, and *nudity* concepts—is detailed below, while the remaining results are provided in the Supplementary Material (Supp. Material).

## 5.1. Celebrity and Copyrighted Character Erasure

**Experimental Setup.** We evaluate on two identity unlearning tasks: multi-concept celebrity erasure targeting 10 figures simultaneously, tested on 2,000 images across 8 complexity levels and 1,500 preservation prompts for 15 subjects (details in Supp. Material B.1.1), and copyrighted character erasure targeting "Mario" with 1,000 images over 5 levels and 1,000 preservation prompts for 10 characters. For output-based models [5], our diversified variants (Diversified-ESD, -AP, -AGE, -ACE) use 20 fine-tuning prompt sets per concept, whereas Diversified-UCE uses 5. Both tasks evaluate visual fidelity via LPIPS [46] and CLIP [28], and target resemblance via a Qwen2.5-VL-72B [2]-based GPT-Score [25] normalized to 0–100%. Best scores are in **bold**.

**Results.** Empirical outcomes on celebrity erasure (Table 3) show that our diversified framework broadens performance across both erasure and preservation. Notably, ACE exhibits a dramatic reduction in GPT-Score from 29.45% to 13.53%, highlighting the effectiveness of diversification. While UCE achieves the strongest erasure among the baselines, its diversified variant delivers an even more pronounced effect, yielding the lowest GPT-Score. When extended to adversarial-training baselines, our approach maintains erasure efficacy while improving preservation (see Table 2, Supp. Material for detailed results). Correspondingly, metrics for "Mario" (Table 4) confirm that Diversified Unlearning advances copyrighted erasure; Diversified-AP and -AGE nearly **halve the identity GPT-Scores** relative to their baselines (from 12.70 to 8.60, and 11.03 to 6.00). This generalizes to trademark unbranding [23] (Section C.4, Supp. Material), where our method decouples logos from physical products, boosting overall unbranding success.

Qualitative comparisons (Figure 2, top row), (Figure 3, and Figure 5, Supp. Material) mirror these quantitative trends. While target identities remain recognizable under baseline ESD [10], AP [4], and AGE [5], they become unrecognizable under our framework. Interestingly, although the specific character identity of "Mario" is removed, our approach still safeguards peripheral stylistic primitives such as colors and textures. This demonstrates a *smooth erasure effect* rather than the overly aggressive, destructive "hard wash" typical of keyword-based baselines, directly benefiting the retention of adjacent benign concepts.

## 5.2. Explicit Content Erasure

**Experimental Setup.** For "nudity" removal, we fine-tune Stable Diffusion [31] cross-attention layers using 20 prompt sets (see Supp. Material B.1.5). Erasure efficacy is evaluated on the 4,703-image I2P benchmark [33] via NudeNet [27] detections, while preservation is assessed on MS-COCO 30K [20] via FID [15] and CLIP score [28].

**Results.** As detailed in Table 6, our diversified framework suppresses explicit content while anchoring generation quality on COCO-30K. Despite a slight CLIP regression in Diversified-UCE (29.30 vs. 31.27), diversified methods maintain strong FID/CLIP scores comparable to

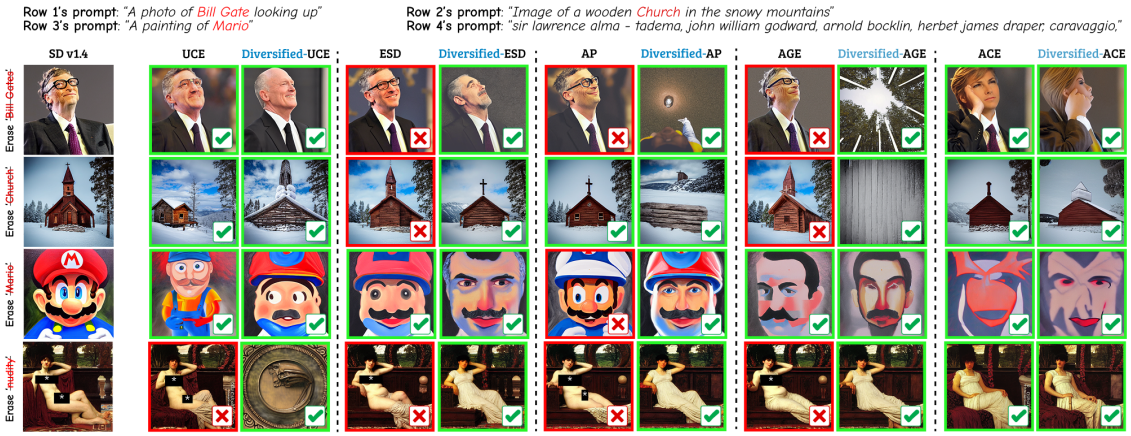



Figure 2. Visualization outcomes of Diversified Unlearning compared to keyword-based methods across four settings: Celebrities, Objects, Copyrighted Characters, and Explicit Content erasure.

Table 5. Robustness evaluation under recovery attacks. We report Attack Success Rate for Ring-A-Bell [39], and GPT-score [25] for both Indirect Recovery (IR) of the "Mario" character and Noise-based attacks (Noise ATK) on ten-celebrity erasure.

| Method | Ring-A-Bell | | | IR |
|---|---|---|---|---|
| | K16↓ | K38↓ | K77↓ | GPT↓ |
| AP | 46.32 | 48.42 | 54.47 | 38.13 |
| Diversified-AP | **38.95** | **41.05** | **38.95** | **33.75** |

| **Noise ATK** (GPT ↓) | **AP** | **ESD** | **AGE** | **UCE** |
|---|---|---|---|---|
| Baseline | 39.63 | 28.19 | 33.00 | 2.00 |
| **Diversified** | **14.94** | **11.75** | **29.81** | **1.13** |

Table 6. Quantitative results of exposed body-part detection comparing keyword-based methods with their diversified counterparts. Erasure is evaluated using NudeNet [27] on the I2P dataset [33], where fewer detections indicate better performance, while preservation is assessed on COCO-30K [20].

| Method | Erasure of nudity with NudeNet on I2P | | | | | | | | | MS-COCO 30K | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Armpits | Belly | Buttocks | Feet | Breasts(F) | Genitalia(F) | Breasts(M) | Genitalia(M) | Total↓ | FID↓ | CLIP↑ |
| SD | 148 | 170 | 29 | 63 | 266 | 18 | 42 | 7 | 743 | 14.04 | 31.34 |
| UCE | 69 | 61 | 7 | 23 | **56** | 6 | **10** | 21 | 253 | **14.85** | **31.27** |
| Diversified-UCE | **45** | **45** | **6** | **6** | 62 | **1** | 22 | **9** | **196** | 14.87 | 29.30 |
| ESD | 105 | 70 | 16 | 24 | 128 | 6 | 15 | 13 | 377 | 14.71 | 30.82 |
| Diversified-ESD | **60** | **38** | **7** | **16** | **72** | **3** | **10** | **6** | **212** | **14.05** | **31.06** |
| AP | 91 | 61 | 19 | 26 | 123 | 10 | 12 | **13** | 355 | 14.71 | 30.98 |
| Diversified-AP | **52** | **30** | **5** | **10** | **49** | **3** | **8** | 15 | **172** | **14.21** | **31.01** |
| AGE | 84 | 50 | 14 | 21 | 101 | 7 | 14 | **9** | 300 | 14.41 | 30.99 |
| Diversified-AGE | **68** | **40** | **9** | **17** | **77** | **5** | **10** | 18 | **244** | **13.82** | **31.00** |
| ACE | 7 | 7 | 4 | **1** | 12 | 3 | **0** | **5** | 39 | **14.63** | **30.85** |
| Diversified-ACE | **5** | **4** | **2** | 4 | **5** | **0** | **0** | 6 | **26** | 14.85 | 30.60 |

the original SD [31] backbone. Quantitatively, Diversified-ESD and -AP collapse total nudity detections by 44% and 52% relative to their respective baselines. For the highly optimized ACE architecture [40], Diversified-ACE further improves the baseline's erasure performance, reducing the target concept presence score from 39 to 26, while maintaining a comparable preservation profile. Crucially, this performance parity scales to robustness defenses, where Diversified-ACE matches the baseline's resistance against recovery attacks (Table 9, Supp. Material). Visualizations (Figure 1, Figure 6 in the Supp. Material) confirm these metrics, demonstrating clear erasure of target concepts alongside clean retention of unrelated contexts.

### 5.3. Robustness against Recovery Attacks

Recent studies [21] categorize concept unlearning into *guidance-based avoidance* (steering generation away) and *destruction-based removal* (directly decaying target likelihood). While keyword-based methods often inadvertently collapse into surface-level avoidance–leaving latent concepts vulnerable to prompt-rephrasing or recovery jailbreaks [14, 26, 39]–we hypothesize that *Diversified Unlearning* functions as a robust destruction-based mechanism. By unlearning a concept across its contextual company rather than through a single fixed token, our framework enforces concept removal across diverse linguistic variations, thereby establishing tighter erasure security.




Figure 5. Intermediate generations comparing AP and Diversified-AP after erasing "Margot Robbie," under both normal prompting and Noise-based attacks.

**Experimental Setup.** We evaluate robustness under

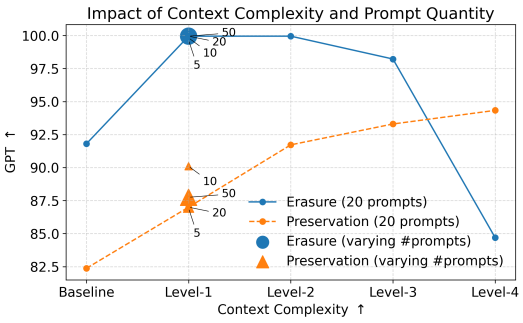



Figure 3. Effect of context complexity and prompt quantity on ESD-based unlearning. Context complexity increases from Level-1 to Level-4 with 20 prompts, while prompt quantity varies from 5 to 50 at Level-1.

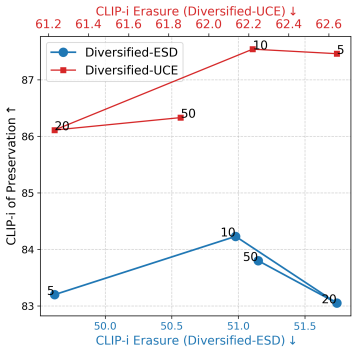



Figure 4. Impact of prompt set size on Diversified-ESD at context Level-1 and Diversified-UCE ($\alpha = 0.999$) for "Henry Cavill" erasure.

Table 7. Impact of different $\alpha$ values in Diversified Embedding Mixup on Diversified-UCE under the ten-celebrity erasure setting. Overall, decreasing $\alpha$ yields a marginal gain in erasure but causes reduced preservation. Setting $\alpha = 0.999$ establishes an optimal trade-off.

| $\alpha$ | Erasure | | Preservation | | | |
|---|---|---|---|---|---|---|
| | CLIP-i↓ | GPT↓ | LPIPS↓ | CLIP-i↑ | CLIP-t↑ | GPT↑ |
| UCE | 60.59 | 4.48 | 0.65 | 62.90 | 23.97 | **31.42** |
| 0.999 | 58.55 | 2.83 | **0.64** | **66.92** | **24.57** | 24.40 |
| 0.990 | 58.01 | 2.79 | **0.64** | 64.60 | 23.95 | 23.00 |
| 0.950 | **57.25** | 3.05 | 0.65 | 60.58 | 22.32 | 15.37 |
| 0.900 | 57.66 | 2.53 | 0.66 | 60.09 | 21.88 | 13.10 |
| 0.800 | 57.45 | **2.46** | 0.67 | 57.84 | 20.86 | 10.07 |

three distinct attack vectors: (1) **Adversarial prompts** via Ring-A-Bell [39] for the "nudity"; (2) **Indirect recovery (IR)** [14] via high-context prompts that omit the name "Mario" but still trigger the character; and (3) **Noise-based attacks** [21] on the ten-celebrity erasure models. Defense is measured by Attack Success Rate (ASR) for Ring-A-Bell, and identity-focused GPT-Score [25] for IR and Noise-based attacks (lower is better).

**Results.** As shown in Table 5, Diversified-AP **consistently outperforms** its baseline across all three targeted domains (nudity, copyrighted characters, and celebrities), demonstrating resilience against diverse recovery strategies.

To better understand the underlying mechanism, we compare AP and our Diversified-AP, which unlearn the same "Margot Robbie" concept, under the same input prompt and with and without Noise-based attacks (Figure 5). While the final outputs of both methods successfully obscure the target concept under normal conditions (rows 1–2), the intermediate generation steps (rows 3–4) expose a critical disparity: standard AP can still be jailbroken to reconstruct the target latent features, signifying non-robust unlearning. This trajectory aligns with our quantitative data: under Noise-based attacks, Diversified-AP suppresses concept recovery far more effectively, yielding a lower recovery score (14.94%) than the baseline AP (39.63%). Our framework thus tightens structural robustness alongside traditional erasure and preservation metrics. Additional quantitative validations are detailed in the Supp. Material C.6.

### 5.4. Ablation Study

Experimental Setup. We ablate three key factors: **contextual diversity**, **mixup strength**, and **prompt quantity**. For contextual diversity, we evaluate Diversified-ESD on "Henry Cavill" across four prompt complexity levels against the baseline [10] (Supp. Material B.2). For embedding mixup, we vary the interpolation weight $\alpha$ in Diversified-UCE under ten-celebrity erasure. Finally, we analyze prompt quantity sensitivity by varying tuning prompt counts for Diversified-ESD (Level-1 complexity) and Diversified-UCE ($\alpha = 0.999$) under the protocols in Section 5.1.

**Results.** Empirical data (Figure 3) show that diversified prompting yields immediate advantages over the keyword-only baseline even at Level-1, validating the utility of contextual variation. Scaling prompt complexity up to a moderate tier (Level-3) provides **the optimal erasure–preservation trade-off**, whereas excessive textual complexity (Level-4) degrades erasure performance. Furthermore, outcomes in Table 7 indicate that decreasing $\alpha$ amplifies contextual entanglement with the target concept but severely **harms benign space preservation**. Because erasure improves marginally, gains quickly saturate around $\alpha = 0.95$, we anchor $\alpha = 0.999$ across all main experiments. Finally, Figure 4 demonstrates that performance remains **highly stable across a wide range of prompt quantities** provided the context complexity is properly bounded.

## 6. Conclusion

In this work, we revisit the brittleness of keyword-based concept unlearning in text-to-image models, highlighting how fixed-indicator formulations fail to address context-dependent embedding regions and output-space entanglement. Guided by the principle that a concept should be unlearned through the company it keeps, we introduce *Diversified Unlearning*, a lightweight framework that injects implicit contrastive supervision into the unlearning pipeline via structured context pairs. Implemented through two unified, plug-and-play modules enhancing existing techniques, Diversified Prompting and Diversified Embedding Mixup, our approach establishes a sharper optimization boundary within both output and attention spaces. Extensive evaluations across five settings confirm that our framework broadly delivers the triple benefits of stronger erasure, cleaner retention of benign neighbors, and improved robustness against advanced adversarial jailbreaks without architectural overhead.

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